A SEQUENCE WHOSE JACOBIAN DETERMINANT IS A UNIT CUTS OUT A REGULAR LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Regular local ring, its embedding dimension and the criterion for a quotient by a sequence in terms of the cotangent space. the author cites Algebra, Lemma algebra-lemma-regular-ring-CM at proof line 11828. Supplies the reduction of the statement to linear independence in the cotangent space.
- Derivations of a ring, the Leibniz rule and the fact that a derivation maps the square of the maximal ideal into the maximal ideal. used at proof lines 11832-11836. Is what makes the linear relation produce a contradiction against the determinant.
- Invertibility of the determinant of the matrix of derivatives of the sequence. the hypothesis of the statement, used at proof lines 11838-11841. Is the condition that the contradiction violates.

Source reference. Original lemma 0GEE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20474–20474. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 868–868. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $(R, \mathfrak{m}, \kappa)$ be a regular local ring. Let $m \geq 1$. Let $f_1, \ldots, f_m \in \mathfrak{m}$. Assume there exist derivations $D_1, \ldots, D_m : R \to R$ such that $\det_{1 \leq i,j \leq m}(D_i(f_j))$ is a unit of R. Then $R/(f_1, \ldots, f_m)$ is regular and $f_1, \ldots, f_m$ is a regular sequence.*

Proof. It suffices to prove that $f_1, \ldots, f_m$ are κ-linearly independent in $\mathfrak{m}/\mathfrak{m}^2$, see Algebra, Lemma 00NQ. However, if there is a nontrivial linear relation then we get $\sum a_i f_i \in \mathfrak{m}^2$ for some $a_i \in R$ but not all $a_i \in \mathfrak{m}$. Observe that $D_i(\mathfrak{m}^2) \subset \mathfrak{m}$ and $D_i(a_j f_j) \equiv a_j D_i(f_j) \bmod \mathfrak{m}$ by the Leibniz rule for derivations. Hence this would imply

$$\sum a_j D_i(f_j) \in \mathfrak{m}$$

which would contradict the assumption on the determinant. □